

Rare Glomerular Disorders

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This chapter describes several uncommon or rare glomerular disorders and the clinical syndromes that tend to affect adults and infrequently have glomerular disease as a major clinical manifestation. These disorders must be recognized and differentiated from other more common glomerular diseases to determine whether a familial disorder is present, estimate the prognosis, plan therapy, or determine the risk for recurrence in a transplanted kidney. The chapter also describes primary and secondary antiphospholipid antibody syndromes in detail.

MESANGIAL PROLIFERATIVE GLOMERULONEPHRITIS WITHOUT IMMUNOGLOBULIN A DEPOSITS

Mesangial proliferative glomerulonephritis (MesPGN) encompasses a heterogeneous collection of disorders with a largely unknown etiology. The common feature is a light microscopy (LM) histologic glomerular pattern of injury characterized by diffuse mesangial proliferation^{1–4} (Fig. 29.1). MesPGN has a diffuse and global increase in mesangial cells, often with an increase in mesangial matrix. Other cells (e.g., neutrophils and monocytes) may contribute to the hypercellularity. Thus, MesPGN is a glomerular pattern of injury, *not a specific disease entity*.

On the other hand, mesangial cellular proliferation can also occur as focal and segmental lesions, often accompanied by segmental glomerular necrosis and very localized crescents. This phenotype may accompany the evolution of pure MesPGN but often signifies the presence of systemic disease, including systemic lupus erythematosus (SLE), immunoglobulin A (IgA) vasculitis (Henoch-Schönlein purpura), IgA nephropathy, infection-related glomerulonephritis (i.e., postinfectious glomerulonephritis [GN] or human immunodeficiency virus [HIV]),⁵ microscopic polyangiitis, granulomatous polyangiitis, Goodpasture disease (anti-glomerular basement membrane [GBM] disease), rheumatoid vasculitis, mixed connective tissue disease, and monoclonal immunoglobulin-associated glomerular diseases (see disease-specific chapters). On occasion, focal and segmental proliferative GN is discovered without any associated identifiable multisystem disease process or IgA deposits (i.e., idiopathic focal and segmental proliferative GN). Such patients have a clinical presentation, course, and response to treatment similar to those described for pure MesPGN but are not discussed further in this section.

In “pure” MesPGN, the peripheral capillary walls are thin and delicate, without obvious deposits, reduplication, focal disruptions, or necrosis. The visceral and parietal epithelial cells, although occasionally enlarged, have not undergone proliferation, and there are no crescents or segments of sclerosis. Large mesangial deposits staining with periodic acid–Schiff (PAS) or fuchsin are absent; such deposits suggest IgA nephropathy (see Chapter 24), lupus nephritis (LN; see Chapter 27), C3 glomerulonephritis (see Chapter 23), or, less often, HIV-associated glomerular diseases (see Chapter 58), which commonly manifest as

a mesangial proliferative lesion. Postinfectious GN may also cause MesPGN (see Chapter 57). The tubulointerstitium and vasculature are usually normal, unless reduced kidney function or hypertension is present or the patient is of advanced age.

On immunofluorescence (IF) microscopy, a wide variety of immunoglobulin and complement deposition is observed (Table 29.1). Most often, diffuse and global granular immunoglobulin M (IgM) and C3 deposits are scattered throughout the mesangium (so-called *IgM nephropathy*; see later discussion), but isolated C3, C1q, or even IgG deposits may be seen.^{6–8} If IgA is the predominant immunoglobulin, the diagnosis is IgA nephropathy. Occasionally, no immunoglobulin deposits are found. Prominent C3 deposits without immunoglobulin deposition should suggest C3 glomerulopathy.⁶ Extensive C1q deposits, with or without immunoglobulin deposits, should suggest C1q nephropathy (see later discussion).⁹ On electron microscopy (EM), the number of mesangial cells is increased, with an occasional leukocyte. Mesangial matrix material is frequently, but not invariably, diffusely increased. Mesangial electron-dense deposits often can be seen but there are no subendothelial, intramembranous, or subepithelial deposits. In the absence of glomeruli for IF, very large mesangial or paramesangial electron-dense deposits suggest IgA nephropathy. Deposits in locations other than the mesangium suggest a postinfectious etiology or an underlying disease, such as LN or C3 glomerulopathy. Deposits of multiple immunoglobulin classes and EM identification of numerous tubuloreticular inclusions suggest underlying LN.

The clinical presentation of MesPGN is varied, but persistent or recurring microscopic or macroscopic hematuria with mild proteinuria is most common. It commonly affects young adults,^{1,2} with a slight male predominance. Nephrotic syndrome is a less frequent initial presentation but more often occurs with diffuse mesangial IgM deposits (IgM nephropathy; see later discussion),³ C1q deposits (C1q nephropathy⁹), or C3 glomerulonephritis.⁶ Pure MesPGN is uncommon (<5%) in patients with idiopathic nephrotic syndrome. In some countries, such as India and China, MesPGN is found in 10% to 15% of kidney biopsy samples, but the frequency is much less in wealthy countries with a lower burden of infectious diseases. MesPGN also has been observed in acute parvovirus B19 disease and in association with Castleman disease.

Kidney function and blood pressure are usually normal, at least initially. Serologic studies are generally unrewarding but should be performed in most patients to exclude known causes. Serum C3 and C4 and hemolytic complement activity (CH50) are typically normal. A low C3 and normal C4 level suggest C3 glomerulopathy or poststreptococcal GN. Antinuclear antibody (ANA), antineutrophil cytoplasmic autoantibody (ANCA), anti-GBM autoantibody, and cryoimmunoglobulins are negative. MesPGN can indicate resolving postinfectious (poststreptococcal or nonpoststreptococcal) GN, often with isolated C3 deposits with scanty, subendothelial, intramembranous, or

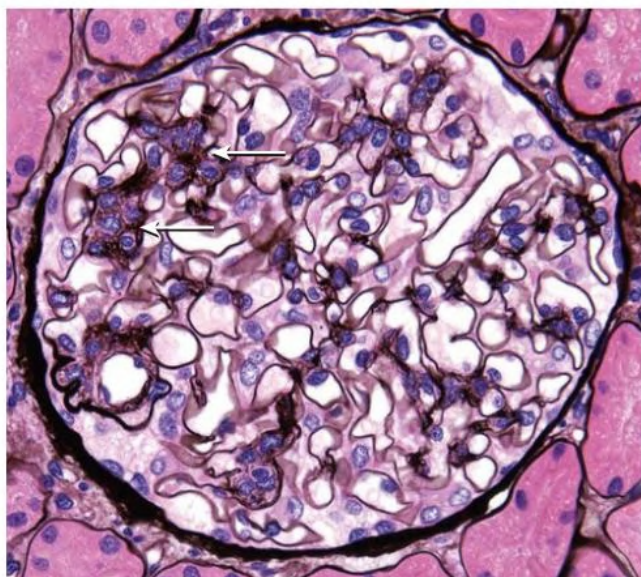


Fig. 29.1 Pure Mesangial Proliferative Glomerulonephritis. There is segmental increase in mesangial cellularity (arrows) with single contoured capillary walls and no sclerosis or epithelial cell hypertrophy.

subepithelial (hump-like) deposits on EM. Urinary protein biomarkers may help determine prognosis.⁷

The prognosis of non-IgA MesPGN depends on the IF findings (see later discussion of IgM and C1q nephropathy) but is often benign. The 30-year kidney survival has been estimated to be 50%.² Severe and persistent nephrotic syndrome indicates a poor prognosis, particularly in males.² Patients with nephrotic syndrome may evolve to typical focal and segmental glomerulosclerosis (FSGS) over time.

The treatment of pure MesPGN, unaccompanied by other underlying diseases or lesions, such as SLE, minimal change disease (MCD) lesion, or IgA nephropathy, is not well defined.^{1,2} The prognosis for patients with isolated hematuria or hematuria combined with mild proteinuria (<500 mg/day) is generally benign, and only supportive measures are needed (see Chapter 82) unless a distinct change in the course is observed. Repeat kidney biopsy may be indicated in such circumstances. For patients with nephrotic syndrome (with or without impaired kidney function), a more aggressive approach is recommended in addition to supportive therapy, especially in the presence of diffuse IgM deposits (IgM nephropathy), as discussed later. Even in the absence of randomized controlled trials (RCTs), an initial course of corticosteroid therapy is justified in most patients with nephrotic-range proteinuria, such as prednisone 60 mg/day or 120 mg every other day for 2 to 3 months, followed by lower doses on an alternate-day regimen for 2 to 3 additional months. About 30% to 50% of these patients experience a decrease in proteinuria to subnephrotic levels, and complete remissions may occur. Relapses of proteinuria are common when steroids are tapered or discontinued. Such relapsing, partially corticosteroid-responsive patients may benefit from the addition of cyclophosphamide, calcineurin inhibitor agents (cyclosporine, tacrolimus, voclosporin), mycophenolate, or rituximab, although information on the efficacy and safety of these agents in pure MesPGN is limited.

Persistent treatment-unresponsive nephrotic syndrome almost invariably progresses to end-stage kidney disease (ESKD), accompanied by a conversion to FSGS over several years. Although transplantation is not contraindicated, patients who rapidly progress to ESKD with superimposed FSGS have a high risk for recurrence of proteinuria and FSGS in the transplanted kidney (see Chapters 19 and 113).

TABLE 29.1 Immunofluorescence Microscopy Patterns in Mesangial Proliferative Glomerulonephritis

Pattern	Associated Disorders
Predominantly mesangial IgA deposits ($\pm$ IgM, C3)	IgA nephropathy
Predominantly mesangial IgG deposits ($\pm$ IgM, C1q, C3)	Often associated with lupus nephritis
Predominantly mesangial IgM deposits ($\pm$ C3)	IgM nephropathy
Mesangial C1q deposits ($\pm$ IgG, IgM, C3)	C1q nephropathy If IgG present, need to exclude lupus nephritis
Isolated mesangial C3 deposits	Often associated with resolving poststreptococcal GN or C3 GN
Negative for immunoglobulin or complement deposits	Idiopathic MesPGN

Ig, Immunoglobulin; GN, glomerulonephritis; MesPGN, mesangial proliferative glomerulonephritis.

Immunoglobulin M Nephropathy

IgM nephropathy is characterized by diffuse and generalized glomerular deposits of IgM often accompanied by C3^{3,8} with mesangial electron-dense deposits. On LM, pure MesPGN is usually observed, sometimes with superimposed FSGS⁸ or crescentic disease. A proposed consensus definition of IgM nephropathy includes (1) dominant mesangial staining for IgM by IF, (2) mesangial electron-dense deposits by EM, and (3) absence of any identifiable systemic disease.⁹ Patients with only IgM deposits by IF but MCD by LM and EM do not have IgM nephropathy, according to this definition. IgM deposits in steroid-sensitive MCD or FSGS do not have any clinical significance.¹⁰ Patients may present with recurring macrohematuria and proteinuria, the latter in the nephrotic range in up to 40% to 50% of patients.⁸ Middle-aged adults are most commonly affected, with a slight male predominance.⁸

Glomerulosclerosis (but not the extent of mesangial proliferation), persisting nephrotic syndrome, and a poor response to corticosteroids or immunosuppressive therapy connote a poor prognosis. As many as 80% of patients with IgM nephropathy and nephrotic syndrome will eventually progress to typical FSGS lesions and, if unresponsive to corticosteroids, will slowly develop progressive chronic kidney disease (CKD). Thus, IgM nephropathy may be an early form of primary FSGS. Patients with IgM deposition accompanying MesPGN but without nephrotic syndrome tend to have a benign course. IgM nephropathy can coexist with IgA nephropathy, diabetic nephropathy, and Fabry disease.^{11–13}

Treatment of IgM nephropathy with nephrotic syndrome is uncertain, although steroid therapy may be associated with a complete or partial remission in as many as 50% of patients. Anecdotal reports of success with rituximab also have appeared. The etiology and pathogenesis are unknown, but IgM deposition itself is believed to be an important pathogenic factor acting to augment complement-mediated injury.

C1q Nephropathy

C1q nephropathy is characterized by IF showing diffuse dominant or codominant mesangial C1q deposition, often accompanied by IgG, IgM, or both,^{4,9} and, much less frequently, by C3 deposits. These immunopathologic features resemble those seen in LN; however, these

BOX 29.1 Rheumatic Diseases Associated With Glomerular Lesions

- Systemic lupus erythematosus (see [Chapter 27](#))
- Rheumatoid arthritis
- Mixed connective tissue disease
- Rheumatic fever
- Ankylosing spondylitis
- Reiter syndrome
- Dermatomyositis/polymyositis
- Scleroderma
- Relapsing polychondritis
- Psoriasis/psoriatic arthritis
- Systemic or renal-limited polyangiitis (see [Chapter 26](#))

patients lack the clinical features of SLE and usually do not develop SLE even after prolonged follow-up. Anti-C1q vasculitis (also known as hypo-complementemic urticarial vasculitis or McDuffie syndrome) can resemble C1q nephropathy, although urticarial lesions and systemic symptoms are more common in anti-C1q vasculitis with low levels of serum C1q and elevated anti-C1q autoantibodies.

LM findings of C1q nephropathy are heterogeneous, including MesPGN (the mesangial proliferative variant), MCD, and FSGS.^{9,14} Electron-dense deposits are seen in mesangial regions often initially near the arteriolar pole and less frequently in subepithelial and subendothelial areas. Nephrotic-range proteinuria occurs, often with hematuria. Males predominate and African Americans are often affected. Serum C3 components, ANA, and anti-double-stranded DNA (anti-dsDNA) antibodies are normal or negative. The response to treatment is poor, and progression to ESKD may occur, particularly when nephrotic syndrome is present, when FSGS lesions predominate, and when the biopsy displays significant tubular atrophy and interstitial fibrosis.^{9,15} Anecdotal reports of success with immunosuppressive agents, including rituximab, have appeared. Patients with C1q deposits and MCD appear to respond well to conventional steroid therapy.

Mesangial Proliferative Glomerulonephritis Associated With Minimal Change Disease

MesPGN may be a part of the spectrum of MCD-FSGS lesions (see [Chapters 18–20](#)). Distinct mesangial hypercellularity superimposed on MCD (diffuse foot process effacement seen on EM) suggests a greater likelihood for the presence of hematuria, corticosteroid unresponsiveness, and an eventual evolution to FSGS.

GLOMERULONEPHRITIS WITH RHEUMATIC, COLLAGEN-VASCULAR, OR CONNECTIVE TISSUE DISEASE

Several collagen vascular diseases other than SLE may be complicated by GN¹⁶ ([Box 29.1](#)), including rheumatoid arthritis (RA), mixed connective tissue disease, polymyositis and dermatomyositis, acute rheumatic fever, scleroderma, relapsing polychondritis, and psoriatic arthritis. IgA nephropathy also may be seen in association with the seronegative spondyloarthropathies. Toxic or hypersensitivity reactions to nonsteroidal antiinflammatory drugs (NSAIDs) can contribute to glomerular disease.¹⁷

Rheumatoid Arthritis

A wide variety of glomerular, tubulointerstitial, and renovascular lesions may complicate RA ([Box 29.2](#)).¹⁶ Clinical abnormalities, including hematuria, leukocyturia, proteinuria, and reduced kidney function are

BOX 29.2 Kidney Disease in Rheumatoid Arthritis

Glomerular Lesions That May Be Direct Complications of the Disease

- MN
- MesPGN ($\pm$ IgA or IgM deposits)
- Diffuse proliferative GN
- Necrotizing and crescentic GN (rheumatoid vasculitis)
- Amyloidosis (AA type)

Glomerular Lesions Associated With Agents Used to Treat Rheumatoid Arthritis

- Gold: MN, MCD, acute tubular necrosis
- Penicillamine: MN, crescentic GN, MCD
- NSAIDs: acute TIN with MCD, acute tubular necrosis, MCD without TIN
- Cyclosporine: chronic vasculopathy and TIN, focal and segmental glomerulosclerosis
- Azathioprine/6-mercaptopurine: TIN
- Pamidronate: collapsing glomerulopathy
- TNF- α inhibitors: lupus-like lesions, crescentic GN

GN, Glomerulonephritis; MCD, minimal change disease; MesPGN, mesangial proliferative glomerulonephritis; MN, membranous nephropathy; NSAIDs, nonsteroidal antiinflammatory drugs; TIN, tubulointerstitial nephritis; TNF, tumor necrosis factor.

common in patients with severe or longstanding RA. Membranous nephropathy (MN; see [Chapter 21](#)) is the most common glomerular lesion. Possibly because of the underlying disease, the presence of human leukocyte antigen (HLA)-DR3 increases the risk for MN in a patient with RA, which is not associated with anti-phospholipase A₂ receptor (anti-PLA₂R) autoantibodies. RA-associated MN has a course similar to that of idiopathic disease, although spontaneous remissions are less likely to occur. Secondary (AA) amyloidosis (see [Chapter 28](#)) is found in up to 30% of patients with RA, of which 56% have proteinuria. Half of these have nephrotic syndrome, which commonly progresses to kidney failure.¹⁸

NSAIDs may produce tubulointerstitial nephritis alone, or with MCD or MN (more commonly the former; see [Chapters 18 and 64](#)). A severe, necrotizing polyangiitis may complicate the course of longstanding RA (rheumatoid vasculitis). These patients may have profoundly reduced C3 levels, striking elevation of rheumatoid factors, and marked polyclonal hypergammaglobulinemia. Kidney involvement in rheumatoid vasculitis is now relatively uncommon. Tumor necrosis factor (TNF)- α inhibitors used to treat RA can evoke a picture resembling LN (see [Chapter 27](#)).

Mixed Connective Tissue Disease

Mixed connective tissue disease is characterized by overlapping features of SLE, scleroderma, and polymyositis. Typically, the serum of such patients contains high-titer autoantibodies to extractable nuclear antigens (ribonucleoprotein-extractable nuclear antigen, U1 ribonucleoprotein antigen) and may have low titers of anti-dsDNA antibody. Kidney disease, originally thought to be quite rare, is found in 10% to 50% of patients, most frequently MN and MesPGN. Steroid treatment is generally effective, but some patients exhibit progressive CKD. Patients with severe GN may respond to treatment regimens similar to those used for LN (see [Chapter 27](#)).

Polymyositis and Dermatomyositis

The related collagen vascular diseases polymyositis and dermatomyositis are characterized by inflammatory lesions in muscle with variable

skin lesions and often include Raynaud phenomenon. Patients may have proteinuria and hematuria secondary to MesPGN with IgM deposits. Acute kidney injury (AKI) may rarely supervene when severe muscle injury and myoglobinuria are present. Treatment with steroids may ameliorate the kidney manifestations and improve the muscle and skin manifestations.

Acute Rheumatic Fever

Acute rheumatic fever secondary to a pharyngeal infection with a rheumatogenic strain of group A β -hemolytic streptococci is seldom accompanied by kidney disease (see [Chapter 57](#)). Cutaneous streptococcal infections are never associated with acute rheumatic fever sequelae. Poststreptococcal GN and acute rheumatic fever almost never coexist because of distinct differences between nephritogenic and rheumatogenic strains of streptococci. Nevertheless, MesPGN is rarely associated with acute rheumatic fever. MesPGN usually manifests with hematuria and scant proteinuria and often resolves with appropriate treatment of acute rheumatic fever.

Ankylosing Spondylitis and Reiter Syndrome (Seronegative Spondyloarthropathies)

The seronegative spondyloarthropathies and oligoarticular arthropathies may be associated with mesangial IgA deposition or MesPGN. Clinical manifestations are usually mild and nonprogressive. AA amyloidosis may complicate longstanding ankylosing spondylitis.

Scleroderma (Systemic Sclerosis)

Scleroderma is a heterogeneous disorder of unknown etiology characterized by uncontrolled expansion of connective tissue in the skin and other visceral organs,¹⁹ as well as vascular thickening and narrowing. Clinical manifestations vary from increased connective tissue in localized patches of skin (morphea) to diffuse and generalized disease (systemic sclerosis). The latter leads to skin thickening of the face and hands, telangiectasia, Raynaud phenomenon, tendon friction rubs, and sclerodactyly. Nail beds display a characteristic pattern of blood vessel abnormalities. Visceral involvement causes interstitial pulmonary fibrosis, loss of esophageal and other gastrointestinal motility, restrictive cardiomyopathy, and kidney disease. Limited forms of the disease (CREST syndrome: calcinosis, Raynaud phenomenon, esophageal dysmotility, sclerodactyly, and telangiectasia) occur but are seldom associated with kidney disease. The disorder is more frequent in females, with an onset usually in young adults. Approximately 90% of patients will have a speckled pattern of fluorescent ANA, and 30% will have detectable antibody to topoisomerase I (Scl-70). Anticentromere antibody is strongly associated with the CREST syndrome. Anti-RNA polymerase III, antitopoisomerase, and anticentromere antibodies are associated with more systemic and kidney involvement and a poor prognosis. Rarely, the visceral abnormalities occur without cutaneous lesions (systemic sclerosis *sine* scleroderma).

Kidney involvement in scleroderma ranges from low-grade proteinuria and slight impairment of glomerular filtration rate to a more marked reduction in renal blood flow leading to a greatly elevated filtration fraction secondary to mild MesPGN to severe AKI.^{20,21} The last is referred to as scleroderma renal crisis (SRC), affecting 5% to 10% of patients with scleroderma and consisting of severe (hyperreninemic) hypertension, encephalopathy, systolic and diastolic congestive heart failure, and AKI. Posterior reversible encephalopathy has been reported. Rare cases of SRC may occur without cutaneous features and even with relatively normal blood pressure. These patients are often positive for anti-RNA polymerase III antibody, and complement activation can be observed.²² There is often accompanying

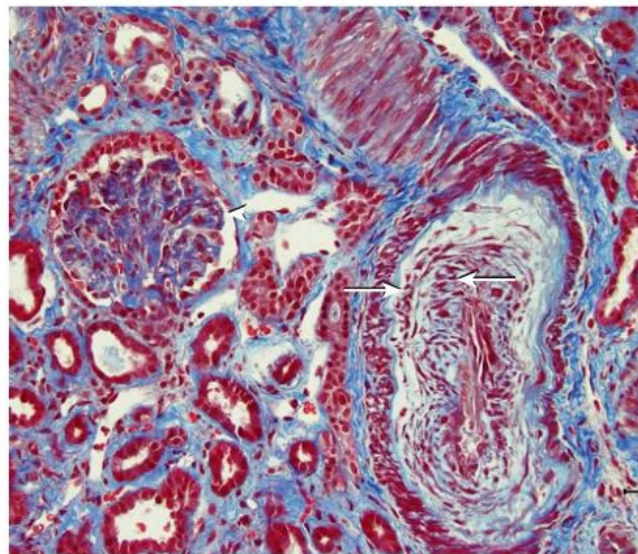


Fig. 29.2 Scleroderma. An interlobular artery shows pronounced mucoid edematous subendothelial thickening with myofibroblasts on Masson trichrome stain (arrows), with endothelial cell swelling and luminal narrowing. The process is limited to the intima with preservation of the internal elastic lamina. Note the ischemic glomerulus (arrowhead) with severely wrinkled and retracted capillary walls.

microangiopathic hemolytic anemia with schistocytes and elevated serum lactate dehydrogenase. AKI results from primary involvement of the arcuate and interlobular arteries ([Fig. 29.2](#)). It may be superimposed on lesions of hypertensive emergencies (e.g., fibrinoid necrosis of the afferent arterioles) and ischemic glomerular changes (e.g., wrinkling of the capillary wall and thickening of the basal lamina).

The prognosis of patients with SRC has improved remarkably with the use of angiotensin-converting enzyme (ACE) inhibitors.^{20,21} In one study, ACE inhibitor treatment was associated with better patient survival at 1 year (75% vs. 15%) and with significant preservation or recovery of kidney function.²² Captopril is the preferred therapeutic agent; there is wide experience with it in SRC and its short half-life facilitates rapid dosing uptitration.²² Various biologic agents, including transforming growth factor (TGF)- β inhibitors, anti-CD20 (rituximab), interleukin 6 (IL-6) monoclonal antibodies, eculizumab, and tyrosine kinase inhibitors are undergoing evaluation as therapies for scleroderma or SRC.²³

In an observational analysis of patients with scleroderma in 19 European kidney registries, the annual incidence of ESKD was 0.11 to 0.26 per 1 million population and the adjusted prevalence was 0.73 to 0.95. Recovery of kidney function occurred in 7.6% (compared with 0.6% for diabetes and 2.1% for nondiabetic kidney diseases). Recovery time may be prolonged. ACE inhibitor treatment should continue during dialysis therapy. Posttransplant 5-year patient survival is 88.2% and allograft survival is 72.4%, but disease in other visceral organs may limit life expectancy.²⁴

Relapsing Polychondritis

Relapsing polychondritis is a chronic, often repetitive, episodic or continuous disorder characterized by destructive inflammation of cartilage (ear, nose, trachea, costal cartilage). Reports have associated it rarely with crescentic GN, MesPGN, MN, or tubulointerstitial IgG4 related disease.²⁵ In one international study of relapsing polychondritis, kidney injury was uncommon and the authors suggested that kidney involvement should raise the possibility of an alternative primary disorder.²⁶

Cartilage lesions may cause deformities (saddle nose, floppy ears, tracheal collapse or stenosis, cardiac valvular disease). When kidney disease occurs, it may be severe and progressive, particularly with crescentic disease. Although corticosteroids may be adequate to treat the nonrenal manifestations, patients with refractory relapsing poly-chondritis may require additional agents such as azathioprine, methotrexate, cyclophosphamide, infliximab, tocilizumab, abatacept, or occasionally rituximab. For progressive kidney disease, cytotoxic agents (e.g., cyclophosphamide) may be indicated, particularly for crescentic GN.

Psoriasis/Psoriatic Arthritis

Psoriasis is an autoimmune inflammatory disease with a strong familial component, with excessive and rapid growth of the skin epidermal layer. Up to 30% of affected patients also develop psoriatic arthritis. Psoriasis (especially if severe or associated with arthritis) is associated with IgA nephropathy, MN, MesPGN, and, very rarely, with membranoproliferative glomerulonephritis (MPGN) pattern of injury on histology.^{27–29} Treatment for psoriasis with glomerular disease may include steroids, methotrexate, calcineurin inhibitors, and TNF- α antagonists, the latter having some success in patients with psoriasis-associated IgA nephropathy.

Non-Lupus “Full House” Glomerulonephritis

Patients can present with findings indicative of an immune-complex mediated glomerulonephritis resembling LN, with deposition of C1q, C3, IgG, IgA, IgM by IF (“full-house”), and a typical MPGN or MN pattern of injury, without the typical clinical and serologic features required for diagnosis of SLE.³⁰ The pathogenesis of this uncommon disorder is unknown, but it may follow certain infections such as COVID-19, parvovirus, HIV, or syphilis.^{31,32} A subset of such patients may evolve to typical SLE over many years. The prognosis seems to be poor,³⁰ and treatment is not well understood, but management similar to that used for LN is often recommended.

Podocyte Infolding Glomerulopathy

This clinicopathologic entity was first recognized in Japan in 2006.³³ It is commonly associated with autoimmune phenomena and connective tissue disease, such as LN or Sjögren syndrome.^{34–36} Proteinuria, including nephrotic syndrome, is universal. The lesion is characterized on EM by microspheres and microtubular structures (likely representing remnants of infolded podocyte cytoplasm) incorporated into thickened GBMs (Fig. 29.3) and can resemble MN or FSGS by LM. There may be concomitant immunoglobulin deposition and subepithelial electron dense deposits in some, but not all, cases.³⁴ The natural history and treatment are largely unknown, but partial remissions of proteinuria can occur.

ANTIPHOSPHOLIPID ANTIBODY SYNDROME

The antiphospholipid antibody syndrome (aPLA syndrome) is characterized by venous and arterial thrombosis and circulating autoantibodies to phospholipid-protein complexes, including those of the coagulation cascade.^{37,38} The syndrome was first recognized by Hughes in 1983³⁷ and has protean manifestations ranging from migraine headaches to multiple thrombosis and multiorgan failure (catastrophic aPLA syndrome [CAPS]).^{37,38} The frequency of thrombotic episodes is about 7.5 per 100 patient years for 5 years after the first thrombotic event and often can be the presenting feature.³⁹ Neurologic symptoms and signs are common, including transient cerebral ischemic attacks (TIAs), strokes, migraine headaches, seizures, myelitis, and balance and sensory disturbances (often resembling multiple sclerosis).^{37,38}

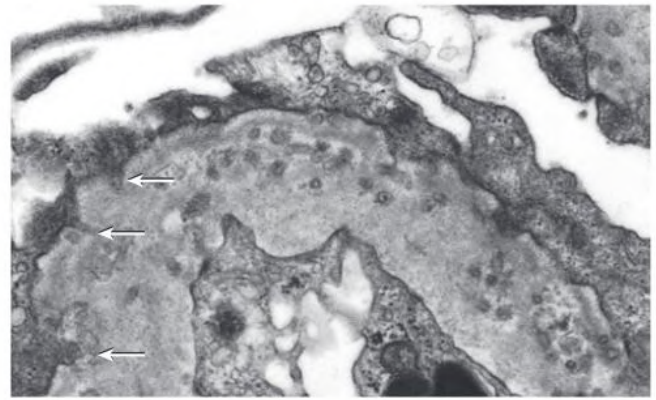


Fig. 29.3 Podocyte Infolding Disease. Electron microscopy shows a thickened glomerular capillary basement membrane (GBM) containing microspheres and microtubular structures. These likely represent entrapped podocyte cytoplasm and, in areas, podocyte cytoplasm is extending into the GBM (arrows).

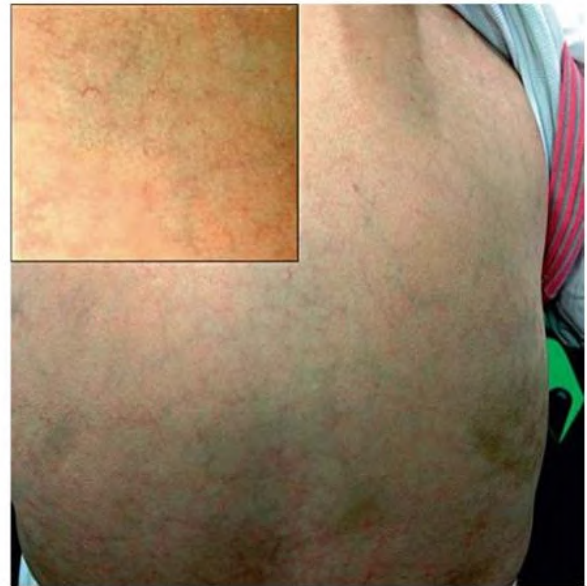


Fig. 29.4 Livedo Reticularis in Patient with Antiphospholipid Antibody Syndrome. Note reticulated skin changes (inset, higher magnification) on the patient's back, in addition to hematomas related to her warfarin therapy. (Courtesy J. Floege, Aachen, Germany.)

Cardiovascular problems, such as pulmonary hypertension, premature atheromatous disease, renal artery stenosis, and myocardial infarction, are common.^{37,38} Livedo reticularis is an important diagnostic clue (Fig. 29.4). Adrenal infarction or hepatic venous thrombosis may lead to acute adrenal insufficiency or Budd-Chiari syndrome, respectively. Visual loss, visual field defects, anosmia, aseptic bone necrosis, fracture, spinal claudication, and autonomic dystrophy are less frequent complications. Pulmonary hemorrhage or fibrosing alveolitis can be a presenting feature.⁴⁰ Repeated pregnancy loss is common.

The kidneys are frequently involved with a thrombotic microangiopathy (TMA; see Chapter 30).⁴¹ CAPS is a rare but often fatal form of aPLA syndrome frequently associated with SLE or infections. It involves many organ systems, including the brain, kidneys, heart, and lungs.⁴²

aPLA syndrome may occur without known systemic disease or may accompany SLE (see Chapter 27) or infection with SARS-CoV-2 (see Chapter 59). In the latter, antibodies are directed at different

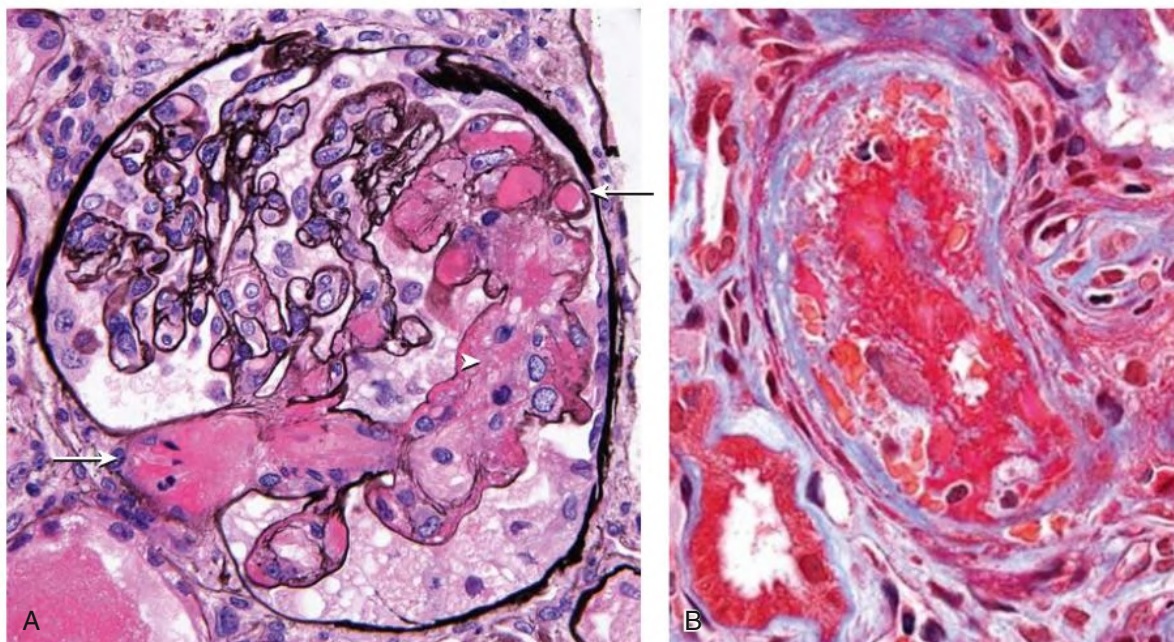


Fig. 29.5 Antiphospholipid Syndrome. (A) Thrombosis of the peri- and intraglomerular arteriole and segmental glomerular capillaries (arrows) with associated mesangiolysis characterized by mesangial matrix dissolution (arrowhead). The adjacent glomerular capillaries show ischemic wrinkling and retraction. (B) Arteriole with luminal fibrin-platelet thrombus and endothelial cell loss. Note the muscular wall is intact and without inflammation.

β_2 -glycoprotein 1 epitopes than in SLE or the primary syndrome, and titers correlated poorly with clinically significant thrombotic events.⁴³ The presence of aPLA syndrome in SLE confers a much worse prognosis and greater risk for neuropsychiatric and cardiovascular complications. aPLA syndrome (primary or SLE related) should always be suspected with a history of migraine headache, TIA or stroke, multiple pregnancy loss, arterial or venous thrombosis, or a family history of autoimmune disease. Diagnosis of aPLA syndrome requires at least one clinical criterion plus one laboratory criterion.⁴⁴ Clinical criteria are (1) thrombosis (arterial or venous) or (2) spontaneous abortions or stillbirths. Laboratory criteria (two or more occasions at least 6–12 weeks apart)⁴⁴ are (1) lupus anticoagulant, (2) anticardiolipin antibody, or (3) anti- β_2 glycoprotein 1 antibody. The pathogenic antibodies in aPLA syndrome appear to be primarily directed against the domain I of β_2 -glycoprotein 1.⁴⁵ Antibodies to prothrombin, thrombin, or phosphatidylserine may predict severe thrombophilia and/or pregnancy loss.

The pathogenesis of aPLA is probably multifactorial.⁴⁶ The thrombotic state seems to involve generation of reactive oxygen species leading to alterations in β_2 -glycoprotein 1 function, impaired function of endothelial nitric oxide synthase, activation of prothrombotic receptors by autoantibodies, increased expression/activation of tissue factor, increased modified forms of prothrombotic factor XI, disruption of annexin A5 shield, and antibody-mediated activation of C3 and/or C5.⁴⁶ There is prominent involvement of the mammalian target of rapamycin complex (mTOR) in aPLA syndrome.⁴⁷ Transplanted patients with aPLA syndrome who receive the mTOR inhibitor sirolimus for prevention of allograft rejection have a lower risk for recurrence of vascular lesions. Complement and toll-like receptor (TLR) activation have also been reported.

Laboratory testing usually reveals an autoantibody to phospholipids (anticardiolipin, anti- β_2 -glycoprotein 1, or prothrombin), but “antibody-negative” aPLA syndrome has been described.^{42,45} False-positive test results for syphilis and a lupus anticoagulant are common.

A prolonged prothrombin or partial thromboplastin time that does not correct when plasma is diluted 1:1 with normal plasma is found in such circumstances. Antiphospholipid antibodies may cross-react with platelet factor 4–heparin complex and thus may generate a false positive antibody test in heparin-induced thrombocytopenia. Mild thrombocytopenia is common (platelet counts about 100,000/mm³ but usually not less than 80,000/mm³). Thrombocytopenia may be an in vitro phenomenon related to the effect of the aPLA antibody on platelet membrane biology. Lymphocytopenia is common in patients with SLE and aPLA syndrome. A mild hemolytic anemia (Coombs negative or positive) may coexist, giving rise to confusion with thrombotic thrombocytopenic purpura, hemolytic uremic syndrome, and Evans syndrome. A frank microangiopathic hemolytic anemia is relatively uncommon. Patients with aPLA syndrome and reduced ADAMTS13 activity may be at higher risk for thrombotic events.⁴⁸ Anti-C1q autoantibodies are found frequently in primary aPLA syndrome and might contribute to abnormal complement activation, particularly in severe refractory cases.

Kidney manifestations of aPLA syndrome are broad and may include renal artery/vein thrombosis, hypertension, small vessel microangiopathy, including glomerular capillaries, and allograft thrombosis (Fig. 29.5). In primary aPLA syndrome, overt kidney manifestations are generally mild and are frequently absent. Nephrotic syndrome is relatively rare. In aPLA syndrome associated with SLE, the kidney manifestations are determined largely by the underlying glomerular disease severity, but the coexistence of aPLA syndrome contributes to a worse prognosis and extrarenal manifestations (neurologic, cardiovascular, osseous, ophthalmologic, pulmonary, hepatic, visceral, and obstetric; see Chapter 27). A strong association of aPLA with alveolar hemorrhage exists in SLE.⁴⁰

The therapy for primary aPLA syndrome and that accompanying SLE is controversial. Immunosuppressive agents, such as steroids or cytotoxic agents, even when they are used to control SLE, have yielded disappointing results. Symptomatic patients are best treated with

anticoagulation. Aspirin (or clopidogrel) can be used in mild cases. Combinations of aspirin and low-dose warfarin might be effective, but the risk for bleeding is increased. Vitamin K antagonism is the mainstay of treatment in severe cases, with the international normalized ratio (INR) adjusted to a level depending on symptoms; INR between 2.0 and 3.5 may be required. Low-molecular-weight heparin (subcutaneous or intravenous [IV]) is the treatment of choice for pregnancy complicated by aPLA syndrome and is also useful in alleviating migraine headache. Aspirin therapy in pregnancy with an aPLA-like syndrome does not appear to be effective. Long-term use of hydroxychloroquine can be beneficial in patients with primary aPLA syndrome without evidence of SLE.⁴⁹ A meta-analysis showed similar venous thrombosis rates in patients treated with direct-acting oral anticoagulants (DOAC) and warfarin but a higher rate of arterial thrombosis with DOAC treatment.⁵⁰

Sirolimus has been suggested for treatment of aPLA syndrome,⁴⁷ but it may increase the risk for thrombosis in susceptible individuals. High-dose IV IgG (IVIG) can have dramatic beneficial effects, especially in acutely evolving disease associated with SLE. Combinations of plasma exchange and IVIG seem to be beneficial in high-risk pregnancy with aPLA syndrome,⁵¹ and plasma exchange alone can be helpful in CAPS.⁵² Given reported TLR activation, plaquenil, which inhibits TLR-7, has been tried and reportedly reduced aPLA in patients with SLE. None of these novel strategies have been tested in controlled trials.

The benefits of immunomodulating agents such as rituximab have not been adequately evaluated in aPLA syndrome, but preliminary reports are encouraging when used alone or with plasma exchange. In severe, life-threatening disease, combinations of rituximab and plasma exchange should be seriously considered. Intensive plasma exchange (plus immunosuppression) has been used in other circumstances, such as in aPLA syndrome associated with SLE, with variable degrees of success. Monoclonal antibodies to C5 (eculizumab) or to CD20 (rituximab) also have been used to treat CAPS with some success.^{53,54}

Other novel therapeutic approaches to the aPLA syndrome, such as coenzyme Q10, statins, *N*-acetylcysteine, phosphodiesterase inhibitors, and factor XI inhibitors have not been thoroughly tested for safety and efficacy.

OTHER UNCOMMON GLOMERULAR DISORDERS

Fibrillary Glomerulonephritis (DNAJB9 Deposition Disease)

Fibrillary glomerulonephritis (FGN) is encountered in about 0.5% to 1% of native kidney biopsies, most commonly in older adults, usually in the sixth decade of life, with females affected more than males.⁵⁵ Proteinuria (including nephrotic syndrome), hypertension, hematuria, and reduced GFR are typically found at presentation. Autoimmune phenomenon, including RA, Sjögren syndrome, lupus-like disorders, and serologic abnormalities (+ANA, ANCA) are seen in 10% to 30% of cases, but hypocomplementemia is rare. Hematologic or nonhematologic malignancies occur in about 8% of cases and may be coincidental. Concomitant hepatitis C virus (HCV) infection is present in about 15% of cases and a small minority (<10%) may have a paraprotein disorder⁵⁵ (see [Chapters 22 and 28](#)). Familial forms have been described.

On LM, an MPGN or MesPGN pattern is often seen with superimposed crescents in about 25% of cases.^{55,56} IF shows irregular or granular IgG deposition, which occasionally may be coarsely linear, resembling anti-GBM disease, or rarely absent by IF.⁵⁷ The defining characteristic is glomerular fibrillary deposits seen by EM, usually predominantly in the subendothelial and mesangial areas

([Fig. 29.6](#)). These fibrils are nonbranching, haphazardly arranged, 15nM in average diameter (range, 9–25nM), and typically Congo Red negative, although 5% may be Congo Red positive.^{55,56,58} The deposits contain codeposited IgG, usually polytypic, complement, and DNAJB9 (Dnaj homolog subfamily member 9).^{55,56} The finding of DNAJB9 deposition is pathognomonic and now required for a correct diagnosis of FGN, leading to a revised nosology for the disease as DNAJB9 deposition disease.^{55,56,59} Elevated serum levels of the DNAJB9 protein may be found; although the specificity of this finding is high (>95%), the low sensitivity (about 67%) precludes employment as a noninvasive screening or diagnostic test.⁶⁰ Thus, the diagnosis of FGN rests on finding DNAJB9-containing glomerular deposits, and this may preclude the need for an EM search for fibrils. A comparison of monoclonal and nonmonoclonal forms of glomerular disease with fibrillary deposits is presented in [Chapter 28](#) (see [Table 28.1](#)).

FGN tends to be progressive, and full remissions are rare. Progression to kidney failure within 2 to 5 years occurs in over 50% of cases, especially in males or those with superimposed crescentic disease and reduced GFR at presentation. Very rarely, anti-GBM disease may coexist with FGN, but such cases have not been proven to be DNAJB9 positive.⁶¹ Therapy is uncertain, but rituximab is associated with stabilization of GFR,^{55,62} even though proteinuria is unaffected. If HCV infection is present, antiviral therapy likely is indicated (see [Chapter 57](#)). FGN uncommonly recurs in kidney transplants, usually with late onset and an indolent course.⁶³

Lipoprotein Glomerulopathy

Lipoprotein glomerulopathy (LPG) is a kidney lipidosis associated with apolipoprotein gene mutations in an autosomal dominant pattern with incomplete penetrance.^{64,65} LPG may be caused by an abnormality in lipoprotein metabolism and/or a lipoprotein secondary structure conformational change related to molecular thermodynamic instability. Most patients harbor mutations of apolipoprotein E (ApoE), usually a heterozygous E2/E3 or E2/E4 phenotype, but homozygosity for ApoE2 or ApoE3 also has been observed. ApoE3 mutations usually are associated with type III hyperlipoproteinemia.⁶⁶ *APOE*-Sendai and *APOE*-Kyoto are the most frequent variants and frequently are in or around the low-density lipoprotein (LDL) receptor binding region. The disease may be associated with psoriasis and hypertensive emergencies accompanied by TMA. Otherwise, there are no distinctive clinical features. Clinically, there usually is heavy proteinuria with nephrotic syndrome and microhematuria may be present.

Histologically LPG is characterized by intracapillary accumulation of apolipoproteins A, B, and E in the glomeruli (mostly ApoE), leading to greatly expanded lumens filled with PAS-negative, bubbly to foamy lipoprotein pseudothrombi, which may be vaguely lamellated and stain positively with oil red O ([Fig. 29.7](#)). Foam cells are not a typical feature, and when present more often indicate crystal-storing histiocytosis, macrophage activation syndrome, ApoE2 glomerulopathy, or lecithin–cholesterol acyltransferase deficiency (see later discussion).⁶⁷ IF stains for ApoE are strongly positive, but immunoglobulins and complement are absent.^{64,68} EM reveals lipid accumulation in capillary lumens and there may be subendothelial, mesangial, and, rarely, subepithelial lipid deposition.

Treatment with bezafibrate or fenofibrate may be effective and is the initial treatment of choice.⁶⁹ About three-quarters of patients with nephrotic syndrome treated with fibrates will achieve a complete or partial remission. Treatment with heparin-induced extracorporeal lipoprotein precipitation-apheresis systems also can confer a complete remission in some patients.⁷⁰

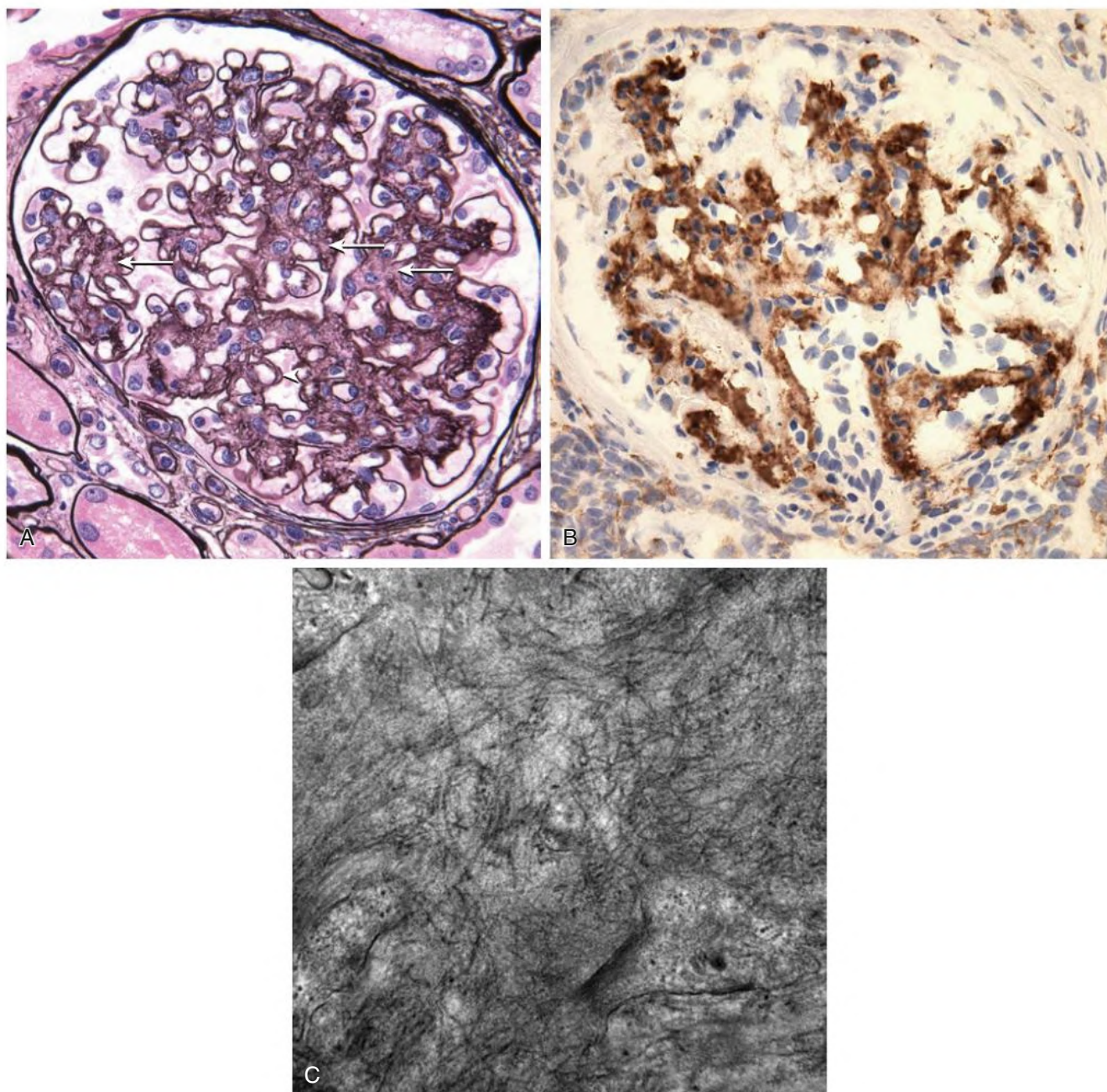


Fig. 29.6 Fibrillary Glomerulonephritis (DNAJB9 Glomerulonephritis). (A) Glomerulus with mesangial expansion because of silver weakly positive to negative material (*arrows*) and rare capillary wall double contours (*arrowhead*). (B) Mesangial staining for DNAJB9 by immunohistochemistry. (C) Electron microscopy shows haphazardly arranged fibrils measuring 13 to 19 nm in the mesangial matrix.

Lecithin–Cholesterol Acyltransferase Deficiency

Lecithin cholesterol acyltransferase (LCAT) is a key enzyme in cholesterol metabolism. It catalyzes the esterification of free cholesterol on the surface of lipoproteins, which is necessary for cholesterol transport. LCAT deficiency is an autosomal recessive disorder associated with two clinical syndromes. Both present with very low plasma high-density lipoprotein (HDL) cholesterol levels but variable and often low frequency of cardiovascular disease in carriers^{71,72} (see [Chapter 20](#)). More than 70 LCAT mutations have been reported causing total LCAT deficiency. This is termed familial LCAT deficiency (FLD), clinically manifesting with corneal opacities (misty deposits, known as “fish eye”), splenomegaly, normocytic normochromic anemia (with target cells), low HDL

cholesterol, ApoA-I and ApoA-II levels, and elevated triglyceride levels. It is likely that glomerular accumulation of oxidized phospholipids and entrapment of an abnormal cholesterol-rich multilamellar particle called lipoprotein-X are pathogenic.⁷³ At least 18 LCAT mutations have been reported to cause the second type of LCAT deficiency called “fish-eye disease” (FED) in which LCAT activity is markedly reduced but not absent. FED patients have reduced plasma HDL and corneal opacities but lack anemia, splenomegaly, or kidney disease.

In FLD, kidney disease is characterized by proteinuria, including nephrotic syndrome, and hypertension. On LM, glomeruli have irregularly thickened, bubbly appearing GBMs, which may have double contours. Mesangial regions are expanded and may have a foamy

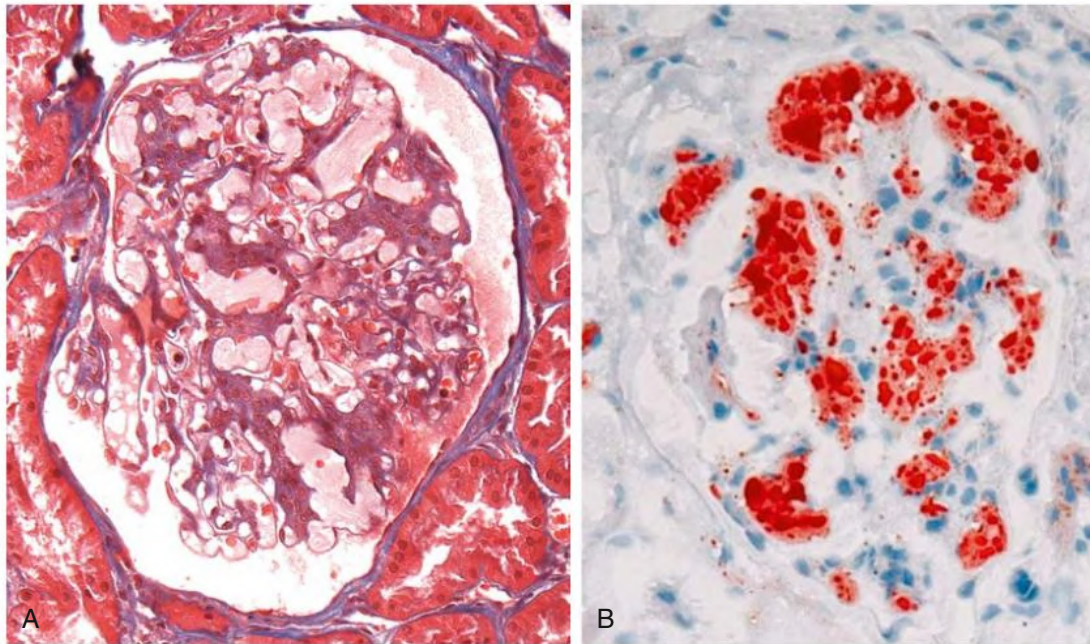


Fig. 29.7 Lipoprotein Glomerulopathy. (A) Dilated capillary lumens contain aggregates of pale, trichrome-stained, mesh-like to granular material. (B) The intraluminal material stains positively with oil red O, confirming its lipid composition.

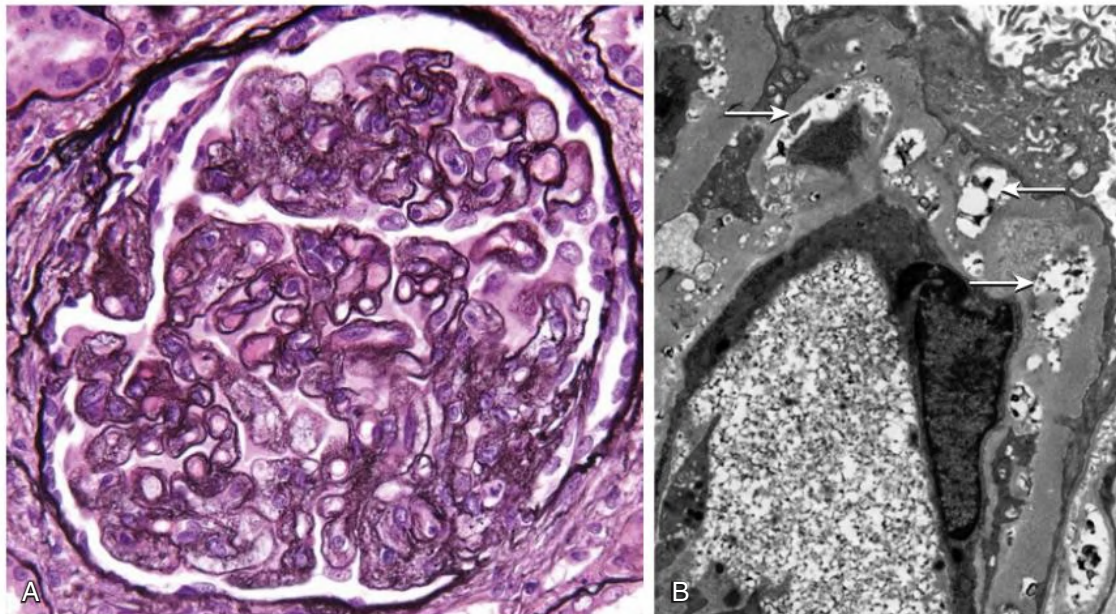


Fig. 29.8 Lecithin-Cholesterol Acyltransferase (LCAT) Deficiency. (A) Irregular, thickened and double contoured glomerular capillary walls containing clear vacuoles, which are characteristic of LCAT deficiency. (B) Electron microscopy reveals electron dense lipid material in electron lucent lacunae (arrows) within the glomerular basement membranes.

appearance with mild hypercellularity. There may be nonspecific granular glomerular staining for C3. Ultrastructurally, the mesangial matrix and GBMs contain electron-dense lipid particles in electron lucent lacunae (Fig. 29.8). Most patients have slow, insidious progressive kidney failure, which is usually detected in the fourth decade of life. In a report of 18 patients followed for a mean of 12 years, 50% of patients developed ESKD by age 46 years.⁷²

Supportive treatment is indicated for proteinuria and hypertension but does not prevent progression. In an LCAT deficiency mouse model,

the HDL cholesterol mimetic CER-001 eliminated circulating and kidney LpX, reducing albuminuria. A clinical trial of this agent is registered for treatment of familial hyperlipidemia and atherosclerotic disease (<https://clinicaltrials.gov/ct2/show/NCT02697136>). Recombinant human LCAT infusions were safe in a phase I clinical trial targeting patients with stable cardiovascular disease (ClinicalTrials.gov NCT01554800). In one patient with LCAT deficiency, it improved anemia and plasma HDL levels and possibly stabilized kidney function.⁷⁴ The disease can recur in kidney transplants, and sequential kidney-liver transplantation might be curative.⁷⁵

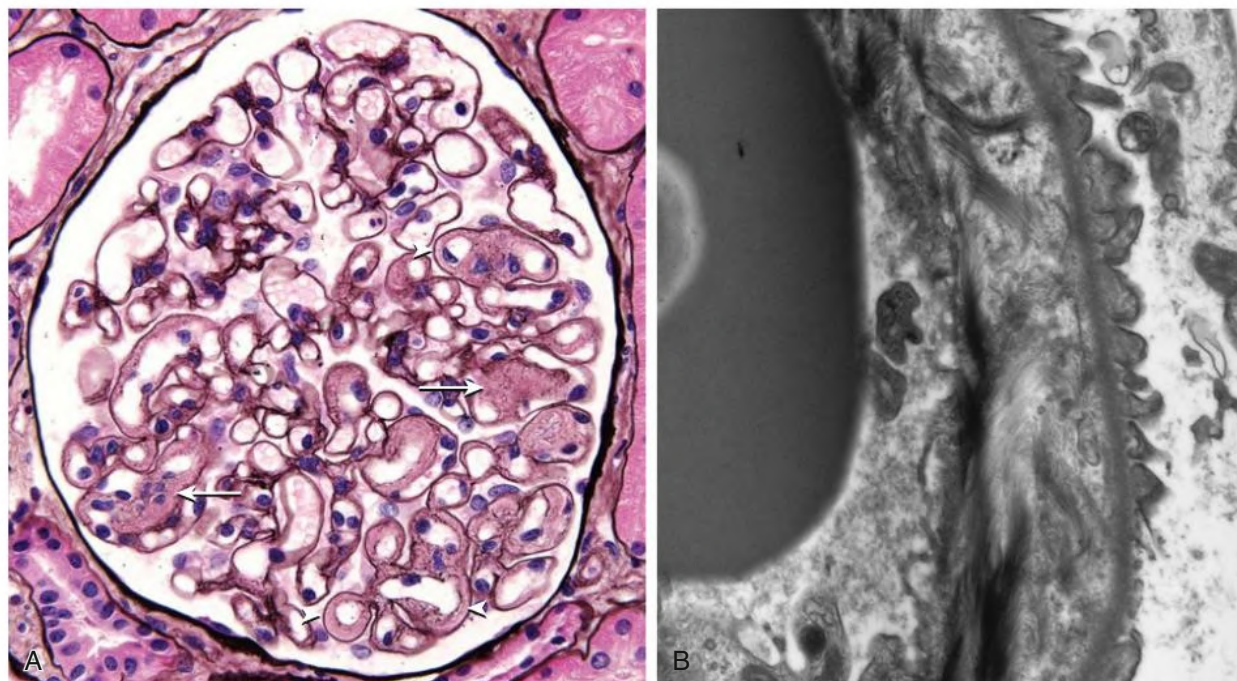


Fig. 29.9 Collagen III (Collagenofibrotic) Glomerulopathy. (A) Mesangial (arrows) and subendothelial (arrowheads) expansion due to silver negative material without hypercellularity. (B) Subendothelial fibrils are randomly oriented with an average 30 nm in diameter and are highlighted with phosphotungstic acid staining.

Collagen III Glomerulopathy

Collagen III glomerulopathy, also known as collagenofibrotic glomerulopathy, is an autosomal recessive systemic disorder with prominent kidney manifestations that may be a forme fruste of nail-patella syndrome (see [Chapter 48](#)), which has similar glomerular features.⁷⁶ However, patients with collagen III glomerulopathy lack the skeletal abnormalities associated with nail-patella syndrome. Although most cases appear to be sporadic, autosomal recessive disease has been reported in children, and *LMX1B* gene mutations are thought to be causative.⁷⁷ Patients present with proteinuria, hypertension, frequent microscopic hematuria, and slowly progressive kidney failure. The disease is more common in Asian populations, patients may be of any age, and there is no sex predilection. Marked elevation of serum hyaluronan concentration and N-terminal procollagen III propeptide may be found and, combined with rare autopsy reports of abnormal type III collagen fibrils seen in liver, spleen, myocardium, and thyroid by EM, suggests that this condition may be because of a systemic defect in type III collagen metabolism.⁷⁸

Kidney biopsy is required for diagnosis. By LM, glomeruli are enlarged with weakly PAS positive, silver negative, Congo red negative material in expanded mesangial regions and widened subendothelial areas ([Fig. 29.9](#)). IF is typically negative or has nonspecific weak staining for immune reactants, although there is strong staining with antisera to collagen type III. EM shows bundles of irregularly arranged, frayed, and curvilinear fibrils showing 60 nm periodicity characteristic of type III collagen,⁷⁹ highlighted with phosphotungstic acid staining.⁸⁰ There is no effective treatment, and there is one report of this disease occurring in a living related kidney transplant.

Fibronectin Glomerulopathy

Fibronectin glomerulopathy is a rare, autosomal dominant, glomerular disease usually presenting in early adolescence with proteinuria, microhematuria, hypertension, distal (type 4) renal tubular acidosis, and slowly progressive kidney failure, reaching ESKD between the

second and sixth decades of life.⁸¹ Mutations occur in the *FN1* gene, which maps to chromosome 2q32.⁸² Kidney pathology demonstrates enlarged, lobular, and normocellular glomeruli with PAS positive, Congo red negative material expanding mesangial and subendothelial areas ([Fig. 29.10](#)). IF is negative or shows low-grade immunoglobulin and complement glomerular staining but is strongly positively with antifibronectin antibody. EM shows usually large finely granular to occasionally fibrillary mesangial and subendothelial deposits with randomly oriented fibrils. The pathogenesis may involve mutations affecting fibronectin tertiary structure, folding, and molecular functionality.⁸³ The differential diagnosis includes other disorders associated with fibril deposition (see [Chapter 28](#)). There is no effective treatment and fibronectin glomerulopathy can recur in kidney allografts.

Nephropathic Cystinosis

Cystinosis is an autosomal recessive lysosomal storage disease with mutations in the cystinosis gene (*CTNS*). Patients with cystinosis often have blond hair, photophobia, hypothyroidism, retinal and corneal deposits, and rickets. They also may develop myopathy, dysphagia, diabetes mellitus, pancreatic exocrine deficiency, pulmonary dysfunction, male hypogonadism, pseudotumor cerebri, and cerebral and vascular calcifications. In the infantile form, Fanconi syndrome starts in the first year of life with tubular proteinuria or nephrogenic diabetes insipidus (see [Chapter 50](#)). Before the advent of cysteamine therapy, proteinuria developed within a few years and ESKD by 10 years of age. Adolescent cystinosis, a variant of pediatric cystinosis with symptom onset into the third decade of life, causes a milder phenotype, although nephrotic syndrome may occur.^{84,85} Treatment of children with cysteamine has resulted in initial presentations with kidney disease in adults as late as age 50 years or beyond. A separate adult form is ocular limited. Cystinosis deficiency results in accumulation of intralysosomal cysteine crystals with subsequent monocyte/macrophage activation. There is damage to glomerular epithelial cells, which

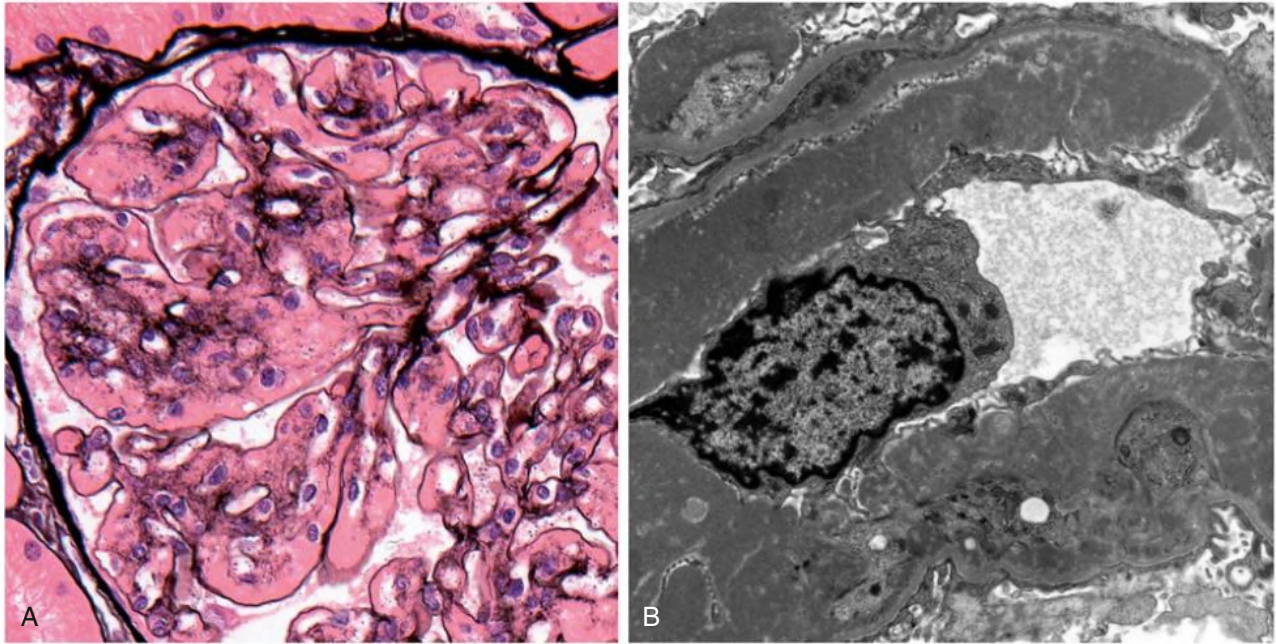


Fig. 29.10 Fibronectin Glomerulopathy. (A) Extensive expansion of mesangial and subendothelial regions due to infiltrating silver negative material with minimal hypercellularity. (B) Electron microscopy shows large subendothelial and adjacent mesangial finely granular deposits.

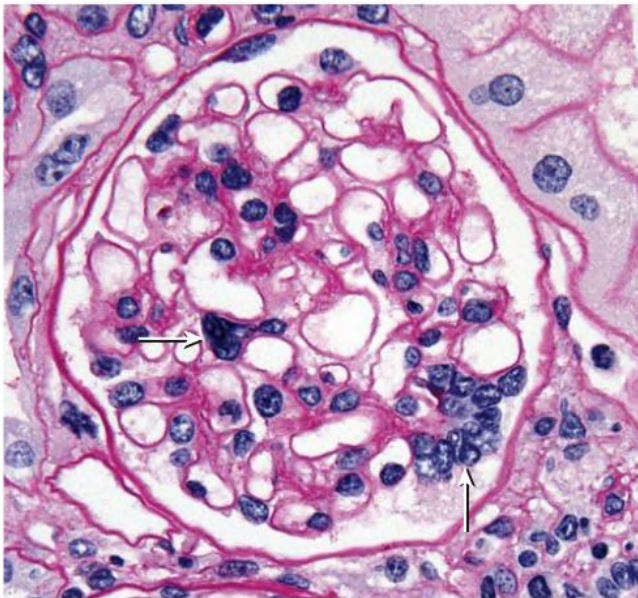


Fig. 29.11 Cystinosis. The glomerulus has segmental multinucleated podocytes (arrows).

become multinucleated, more often seen in the infantile form, with podocyte foot process effacement and progression to FSGS and global glomerulosclerosis^{86,87,88} (Fig. 29.11). Cysteine crystals may be found in glomerular epithelial and endothelial cells, tubular epithelial cells, and infiltrating interstitial macrophages. The treatment is long-term cysteamine administration.⁸⁹

Miscellaneous Storage Diseases Rarely Associated With Glomerular Lesions

Diseases associated with tissue storage of abnormal lipids or carbohydrates may rarely provoke glomerular lesions, usually in infants and children. Glomerular endothelial cell inclusions may be found

in Niemann-Pick disease and neuronal ceroid lipofuscinosis, whereas mesangial cell inclusions occur with gangliosidosis. Diseases with podocyte vacuolization include Hurler syndrome (type I mucopolysaccharidoses), aspartylglucosaminuria, mannosidosis, Refsum disease, nephrosialidosis and I cell disease (mucopolipidosis type II). Infiltrating macrophages containing inclusions may occur in Gaucher disease, and enzyme replacement therapy has been reported to induce complete remission from Gaucher-induced nephrotic syndrome.⁹⁰ FSGS has been reported in those with von Gierke disease (glycogen storage disease). In storage diseases for which a recombinant enzyme treatment is available, controversy exists regarding the relative benefits of enzyme replacement, substrate reduction, and/or hematopoietic stem cell transplantation. The central nervous system (CNS) is not protected by enzyme replacement therapy because of blood-brain barrier impermeability.

Juvenile malabsorption of vitamin B₁₂ with megaloblastic anemia (Imerslund-Grasbeck syndrome; cubulin deficiency) may cause prolonged glomerular proteinuria (albuminuria), but progressive kidney disease does not develop. Patients with cubulin mutations with proteinuria but without megaloblastic anemia have also been reported. Asphyxiating thoracic dystrophy (Jeune syndrome) is associated with glomerular, tubular, and interstitial abnormalities. Hereditary osteolysis causing wrist and ankle arthralgias and deformities can be accompanied by chronic GN.

Idiopathic (Nondiabetic) Nodular Glomerulosclerosis

Nodular glomerulosclerosis, an intercapillary nodular mesangial expansion encroaching on glomerular capillary lumens, is most commonly associated with diabetes mellitus (Kimmelstiel-Wilson lesion) and proliferative diabetic retinopathy (see Chapter 31). Similar LM lesions are reported in patients with impaired glucose tolerance not meeting criteria for diabetes mellitus or may persist after marked weight loss with remission of glycemic abnormalities, masking previous type 2 diabetes. Other diseases potentially associated with a nodular glomerulosclerosis pattern include monoclonal paraproteins (light chain and heavy chain, including Waldenström macroglobulinemia; see Chapter 28);

cystic fibrosis; hepatitis C; cigarette smoking; chronic cyanotic/ischemic syndromes; chronic thrombotic microangiopathy; protein-folding diseases such as amyloidosis, FGN, and immunotactoid GN; collagenofibrotic glomerulopathy; and fibronectin glomerulopathy.

Idiopathic nodular glomerulosclerosis was first described in 1989. Its pathogenesis is unknown, and it is a diagnosis of exclusion. It may be related to smoking, obesity, longstanding hypertension, and intermittent TMA; the term smoking-related glomerulopathy has been suggested.^{91,92} Recent data suggest potential contributions from a mechanism that is common to diabetic and nondiabetic nodular glomerulosclerosis and involves epigenetic changes in the annexin gene, leading to collagen type VI secretion and deposition by glomerular endothelial cells.⁹³

The clinical features are nonspecific and nondiagnostic. The average age of patients with idiopathic nodular glomerulosclerosis is approximately 70 years; most are male and presentation with nephrotic syndrome is common. LM shows nodular glomerulosclerosis identical to the diabetes-associated Kimmelstiel-Wilson lesions, often with neovascularization within the nodules. GBMs are thickened and there may be subendothelial lucencies with capillary double contour formation. There are varying degrees of arteriolonephrosclerosis and hyalinosis. No electron-dense or organized deposits are seen on EM. The GBMs and tubular basement membranes may have pseudolinear staining for IgG and albumin on IF, similar to diabetic kidney disease.

The prognosis is poor and relates to persistence of nephrotic-range proteinuria. Most patients with idiopathic nodular glomerulosclerosis will progress to ESKD, sometimes quite rapidly, with 50% kidney survival 1 year after diagnosis in those who continue to smoke heavily. There is no known therapy other than angiotensin inhibition to reduce the proteinuria and smoking cessation where relevant.

Macrophage Activation Syndrome/Histiocytic Glomerulopathy

Macrophage activation syndrome (MAS) and hematophagocytic lymphohistiocytosis (HLH) are uncommon disorders, both characterized by massive expansion and activation of T-lymphocytes and tissue macrophages with hematophagocytic capacity, leading to overproduction of cytokines and cytokine storm. The bone marrow shows expansion of histiocytes expressing CD163, with active hemophagocytosis. By convention, HLH is a primary disease associated with causative, usually autosomal recessive, mutations. MAS, also known as secondary HLH, presents similarly, but is often associated with an underlying triggering condition. MAS occurs in approximately 15% of patients with systemic juvenile idiopathic arthritis (SJIA) and less frequently in patients with SLE, other autoimmune disorders, neoplasms, and infections (especially EBV, CMV, and SARS CoV-2). Two separate but overlapping criteria for the diagnoses of HLH and MAS are published.^{94,95}

Typically, patients present with pancytopenia, fever, rash, coagulopathy, liver function abnormalities, adenopathy, hepatosplenomegaly, erythrophagocytosis, and/or TMA. Excessive production of cytokines (TNF- α and IL-1, -4, -6, -8, -10), extreme hyperferritinemia, marked hypertriglyceridemia, and hypofibrinogenemia are common findings often leading to the correct diagnosis. The erythrocyte sedimentation rate may be low. A clinical diagnosis should be confirmed by bone marrow biopsy.

Kidney manifestations include AKI and nephrotic syndrome. Collapsing FSGS has been reported. Histiocytic glomerulopathy is more specific to MAS and HLH, characterized by extensive glomerular capillary infiltration with macrophages with or without erythrocyte phagocytosis or foam cells and often associated with endothelial

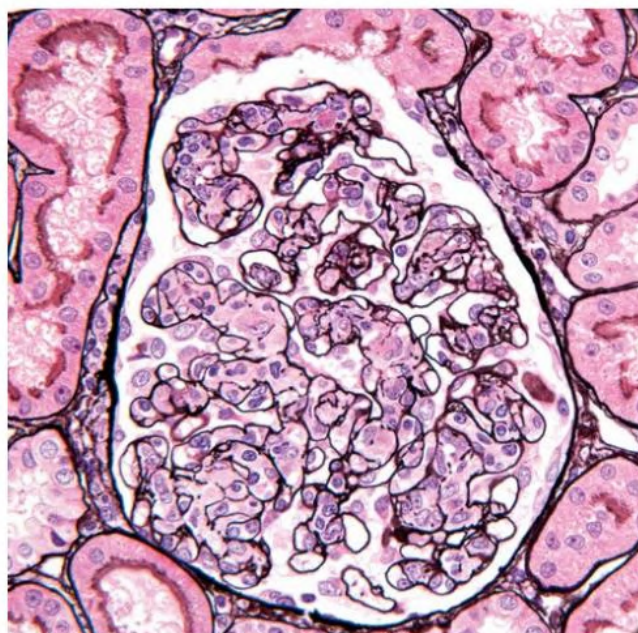


Fig. 29.12 Histiocytic Glomerulopathy. Capillary lumens and less often mesangial regions contain many monocyte/macrophages, which stain positively for CD68 and CD163 in immunohistologic examinations.

cell swelling and features of TMA⁹⁶ (Fig. 29.12). There are no immune deposits by IF.

The mortality rate of MAS is high and treatment is difficult, initially consisting of removing the triggering agent (if possible). High-dose steroids, calcineurin inhibitors, IVIG, TNF- α and IL-6 antagonists, and IL-1 receptor antagonists (anakinra) are the most commonly used therapies. Rituximab, alemtuzumab, and plasma exchange have also been used with varying degrees of success.⁹⁷ In children with a genetic mutation causing primary HLH, bone-marrow transplantation may be lifesaving.⁹⁸

DRESS Syndrome

DRESS (drug reaction with eosinophilia and systemic symptoms) syndrome is an uncommon manifestation of severe drug hypersensitivity characterized by extensive mucocutaneous rashes, fever, lymphadenopathy, hepatitis, eosinophilia, and atypical lymphocytosis with multisystem organ injury.⁹⁹ Reactions to aromatic anticonvulsants, such as phenytoin and carbamazepine, are a common cause but several medications, including antibiotics and allopurinol, have been implicated. AKI and proteinuria can be seen, and kidney biopsy usually reveals acute tubulointerstitial nephritis containing numerous eosinophils. Untreated, DRESS syndrome can be fatal. For severe cases, parenteral and oral steroids are indicated.¹⁰⁰

Kimura Disease

Kimura disease (angiolymphoid hyperplasia with eosinophilia [ALHE]) is a rare disorder characterized by nonmalignant masses in the head and neck, lymphadenopathy, marked eosinophilia, elevation of serum IgE, and immune thrombocytopenia, predominantly in patients of Asian ancestry.¹⁰¹ Kidney involvement with nephrotic syndrome because of MN, FSGS (tip lesion variant), or MesPGN is common. Steroid therapy is successful in most cases, although relapse upon tapering is common. Use of concomitant steroid-sparing agents including *Tripterygium wilfordii*, leflunomide, tacrolimus, vincristine, and mycophenolate have been anecdotally reported.

SELF-ASSESSMENT QUESTIONS

1. A 42-year-old man is discovered to have proteinuria, impaired kidney function (serum creatinine 2.3 mg/dL), and normocytic normochromic anemia (hemoglobin 8.2 g/dL). The physical examination shows hypertension (150/98 mm Hg), mild obesity (body mass index 31 kg/m²), bilateral corneal opacities, and 1+ edema. The albumin is 3.2 g/dL, total cholesterol 120 mg/dL, low-density lipoprotein cholesterol 80 mg/dL, high-density lipoprotein cholesterol 15 mg/dL, and triglycerides 200 mg/dL. Serum C3 is normal. A fasting blood sugar is 120 mg/dL. Urinary total protein excretion is 4.6 g/day. Which of the following is the *most likely* diagnosis?
 - A. Alport syndrome
 - B. Adult-onset cystinosis
 - C. Lecithin–cholesterol acyltransferase deficiency
 - D. C3 glomerulopathy
 - E. Obesity-related glomerulopathy
2. Which of the following can be an effective treatment of lipoprotein glomerulopathy?
 - A. Cyclosporine
 - B. Bezafibrate
 - C. Atorvastatin
 - D. Corticosteroids
 - E. Plasma infusions
3. A 65-year-old woman is found to have nephrotic syndrome, and kidney biopsy shows membranous nephropathy. Serum C3 is normal, and ANA is 1:80. Anti-dsDNA is negative. Immunofluorescence staining of the biopsy reveals extensive deposits of IgG4 and weak deposits of IgG1 and IgG3. Serologic studies for hepatitis B are negative. Which of the following is the *most likely* diagnosis?
 - A. Membranous lupus nephritis
 - B. Membranous nephropathy secondary to cancer
 - C. Idiopathic (primary) membranous nephropathy
 - D. Membranous nephropathy secondary to hepatitis C infection
4. A 46-year-old woman complains of frequent “migraine” headaches. She is found to be anemic (hemoglobin 9.8 g/dL), and serum creatinine is elevated to 1.6 mg/dL. Her blood pressure is 156/94 mm Hg. Urinalysis reveals 2+ protein and a trace of blood. The serum C3 and C4 are normal, and ANA is negative. She takes an NSAID for headache. She has had three miscarriages and is presently amenorrheic. Her physical examination is otherwise unremarkable. There are no localizing neurologic findings. Which of the following tests would be *most* appropriate as a next step in the diagnostic evaluation of this patient?
 - A. Anti-dsDNA autoantibody
 - B. Magnetic resonance imaging of the brain
 - C. Serum iron and iron-binding capacity
 - D. Antiphospholipid antibody test
 - E. Serum protein electrophoresis

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